Classification of Closed Surfaces

In this talk, we assume the existence of triangulations of surfaces, and prove the classification theorem for closed surfaces. In Part I, we develop the theory of polygonal presentations, along with a simple example. In Part II, we show the classical proof made by Seifert and Threlfall (1934), which can be found in most textbooks like GTM202. In Part III, we introduce a new proof by Conway (1992), which avoids tedious combinatorial discussions.

Part I: Definitions and Basic Properties

We define polygonal presentations and outline transformations that preserve the homeomorphism class of the geometric realization.
Given a set S, we denote a **word** W in S as a finite list $W = a_1 a_2 \cdots a_n$, where $a_i = s \text{ or } s^{-1}$ with $s \in S$. A polygonal presentation $\mathcal{P} = \langle S \mid W_1, \cdots, W_n \rangle$ consists of a set S and words W_i in S of length no less than 3 ($\langle a \mid aa \rangle, \langle a \mid aa^{-1} \rangle$ are permitted as a special case).
Any polygonal presentation determines a topological space $|\mathcal{P}|$, called the geometric realization of $\mathcal{P}$, defined as follows:

1. For each word W_i of length k, construct a regular $k-$gon P_i, and mark the k edges using the k letters of W_i bijectively in counterclockwise order.
2. Let $|\mathcal{P}|$ be the quotient space of $\bigsqcup_i W_i$ by pasting edges with the same letter together. Edges marked a are identified preserving orientation, and edges marked a^{-1} are identified reversing orientation.

Consider the following homeomorphism-preserving transformations:

- Reflection: $\langle S \mid a_1 \cdots a_m, W_2, \cdots, W_k \rangle \mapsto \langle S \mid a_m^{-1} \cdots a_1^{-1}, W_2, \cdots, W_k \rangle$.
- Rotation: $\langle S \mid a_1 \cdots a_m, W_2, \cdots, W_k \rangle \mapsto \langle S \mid a_2 \cdots a_m a_1, W_2, \cdots, W_k \rangle$.
- Cutting & Pasting: $\langle S \mid W_1 W_2, W_3, \cdots, W_k \rangle \leftrightarrow \langle S, e \mid W_1 e, e^{-1} W_2, W_3, \cdots, W_k \rangle$. (If W_1, W_2 have length at least 2).
- Folding & Unfolding: $\langle S, e \mid W_1 e e^{-1}, W_2, \cdots, W_k \rangle \leftrightarrow \langle S \mid W_1, \cdots, W_k \rangle$.

Call a pair of edges with the same identifier complementary if they are a, a^{-1} and twisted if they are a, a or a^{-1}, a^{-1} (since the surface twists when one pastes them together).
As an exercise in these operations, we prove a surprising lemma:
Lemma: $K \cong \mathbb{P}^2 \# \mathbb{P}^2$ and $\mathbb{T}^2 \# \mathbb{P}^2 \cong \mathbb{P}^2 \# \mathbb{P}^2 \# \mathbb{P}^2$.
Proof: First show that $K \cong \mathbb{P}^2 \# \mathbb{P}^2$:

$$\begin{aligned} K &= \langle a, b \mid abab^{-1} \rangle \cong \langle a, b, c \mid abc, c^{-1} a b^{-1} \rangle \cong \langle a, b, c \mid bca, a^{-1} cb \rangle \\ &\cong \langle b, c \mid bccb \rangle \cong \langle b \mid bb \rangle \# \langle c \mid cc \rangle = \mathbb{P}^2 \# \mathbb{P}^2. \end{aligned}$$

Next $\mathbb{T}^2 \# \mathbb{P}^2 \cong K \# \mathbb{P}^2$:

$$
\begin{aligned}
K \# \mathbb{P}^2 &= \langle a,b,c \mid abab^{-1}cc\rangle \cong \langle a,b,c,d \mid cabd, d^{-1}ab^{-1}c\rangle \\
&\cong \langle a,b,c,d \mid abdc, c^{-1}ba^{-1}d\rangle \cong \langle a,b,d \mid abdba^{-1}d\rangle \\
&\cong \langle a,b,d,e \mid bde, e^{-1}ba^{-1}da\rangle \cong \langle a,b,d,e \mid deb, b^{-1}ea^{-1}d^{-1}a\rangle \\
&\cong \langle a,d,e \mid deea^{-1}d^{-1}a\rangle \cong \mathbb{T}^2 \# \mathbb{P}^2.
\end{aligned}
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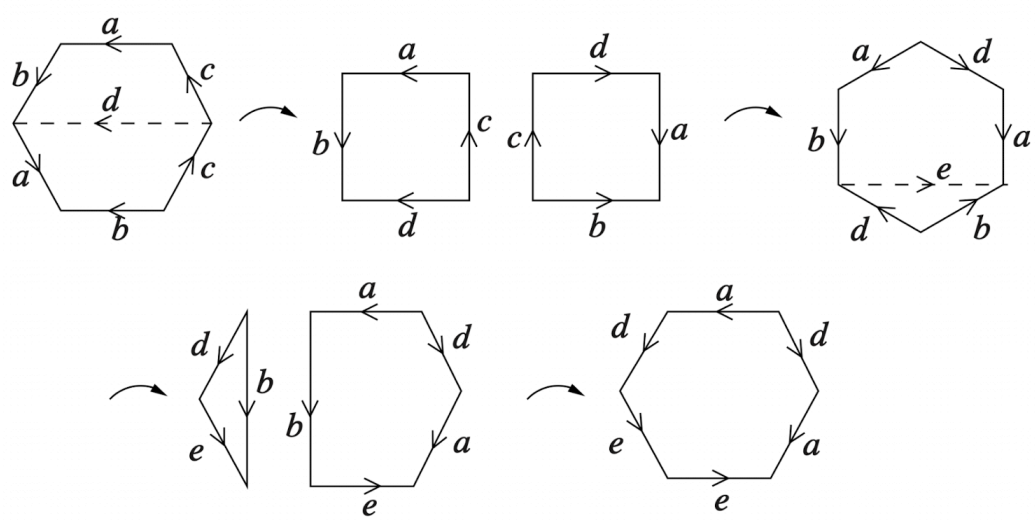

Fig. 6.19: Transforming $\mathbb{P}^2 \# \mathbb{P}^2 \# \mathbb{P}^2$ to $\mathbb{T}^2 \# \mathbb{P}^2$.

Our goal is to simplify any polygonal presentation into disjoint blocks of $aba^{-1}b^{-1}$ (tori) and cc (projective planes). By this Lemma, any mixture of tori and projective planes reduces entirely to connected sums of projective planes.

Part II: Seifert and Threlfall's Proof

The proof goes by devising algorithms to repeatedly apply the preceding transformations, and simplify the presentation step by step.
Suppose M is a compact connected surface, we assume M has a triangulation, then M has a presentation $\mathcal{P} = \langle S \mid W_1, \cdots, W_n\rangle$ where S are the edges of the triangulation, and W_i are the triangles.

Step1: Reduce to a single face

Since M is connected, there must be disjoint faces that share an edge. By repeatedly applying the pasting transformation, we merge all faces into a single polygon.
Suppose M is not the sphere $\mathbb{S}^2 \cong \langle a \mid aa^{-1}\rangle$ in the following steps.

Step2: Eliminate adjacent complementary edges

Trivial since any sequence aa^{-1} can be eliminated by folding.

Step3: Grouping twisted edges

Otherwise if a twisted pair is not adjacent, the word is $VaWa$ where V, W are non-empty. Then

$$\begin{aligned} VaWa &= \langle S, a \mid VaWa \rangle \cong \langle S, a, b \mid Vab, b^{-1}Wa \rangle \cong \langle S, a, b \mid bVa, a^{-1}W^{-1}b \rangle \\ &\cong \langle S, a, b \mid bVW^{-1}b \rangle \cong \langle S, b \mid VW^{-1}bb \rangle. \end{aligned}$$

After this process, the number of nonadjacent pairs decreases by at least one, so after finitely many such operations, there are no nonadjacent twisted pairs. Adjacent complementary pairs that are created can be eliminated by Step2, without introducing new nonadjacent pairs.

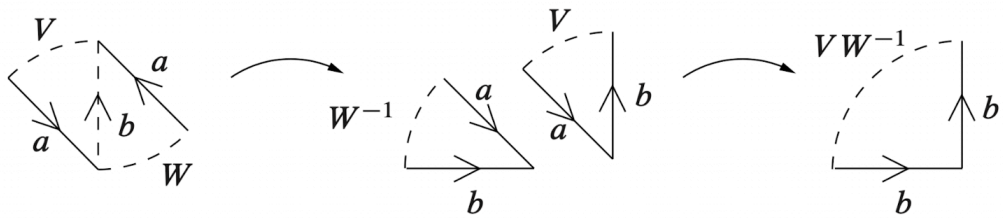

Fig. 6.20: Making a twisted pair adjacent.

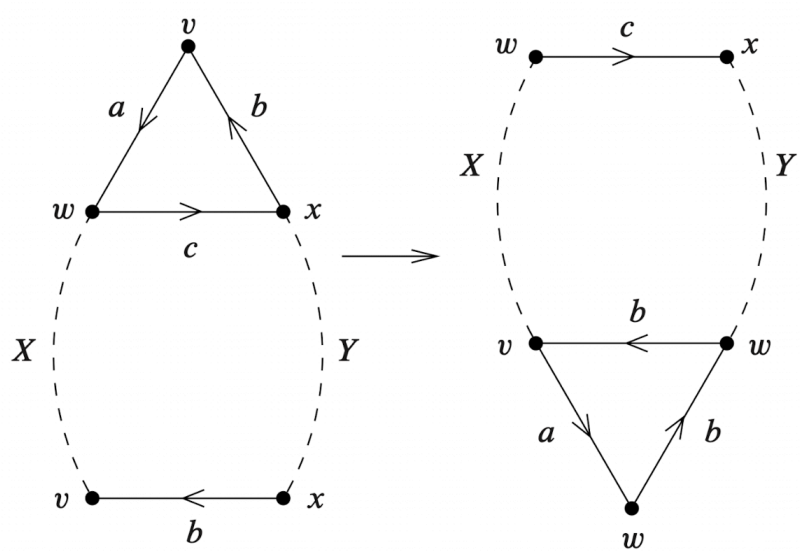

Fig. 6.21: Reducing the number of vertices equivalent to v.

Step4: Merge vertices into a single equivalence class

Suppose the polygon's vertices fall into multiple equivalence classes. There must be an edge a connecting a vertex from class v to a vertex from class $w \neq v$. Let b be the adjacent edge sharing the v vertex. By Step2 $b \neq a^{-1}$ and clearly $b \neq a$. Since b or b^{-1} must reappear, suppose the presentation is $baXb^{-1}Y$ (the case with $baXbY$ is analogous). Note that

$$\begin{aligned} \langle S, a, b \mid baXb^{-1}Y \rangle &\cong \langle S, a, b, c \mid bac, c^{-1}Xb^{-1}Y \rangle \cong \langle S, a, b, c \mid acb, b^{-1}Yc^{-1}X \rangle \\ &\cong \langle S, a, c \mid acYc^{-1}X \rangle. \end{aligned}$$

Since the edge b, b^{-1} both have vertices v, the process decreased the number of $v-$vertices by at least 1, and added one $w-$vertex. Adjacent complementary pairs created can be eliminated by Step2 without increasing the number of $v-$vertices.
Repeating this eliminates the class v entirely, and eventually there will only be one class.

Step5: Complementary pairs must be intertwined

Note that adjacent complementary pairs have been eliminated. Otherwise suppose the presentation is $aXa^{-1}Y$ where X, Y have no edges with common identifiers (since all twisted pairs are adjacent). Then any vertex in X will not be pasted with a vertex in Y, a contradiction to Step4.

Step6: All intertwined pairs are just in the form $aba^{-1}b^{-1}$:

Suppose the presentation is $WaXbYa^{-1}Zb^{-1}$, then

$$\begin{aligned} WaXbYa^{-1}Zb^{-1} &= \langle S, a, b \mid WaXbYa^{-1}Zb^{-1} \rangle \cong \langle S, a, b, c \mid WaXc, c^{-1}bYa^{-1}Zb^{-1} \rangle \\ &\cong \langle S, a, b, c \mid XcWa, a^{-1}Zb^{-1}c^{-1}bY \rangle \cong \langle S, b, c \mid XcWZb^{-1}c^{-1}bY \rangle \\ &\cong \langle S, b, c, d \mid WZb^{-1}d, d^{-1}c^{-1}bYXc \rangle \cong \langle S, b, c, d \mid dWZb^{-1}, bYXcd^{-1}c^{-1} \rangle \\ &\cong \langle S, c, d \mid WZYXcd^{-1}c^{-1}d \rangle. \end{aligned}$$

Likewise this process does not split intertwined pairs that are previously adjacent, and makes a new intertwined pair adjacent. Repeating this process for each intertwined pair.

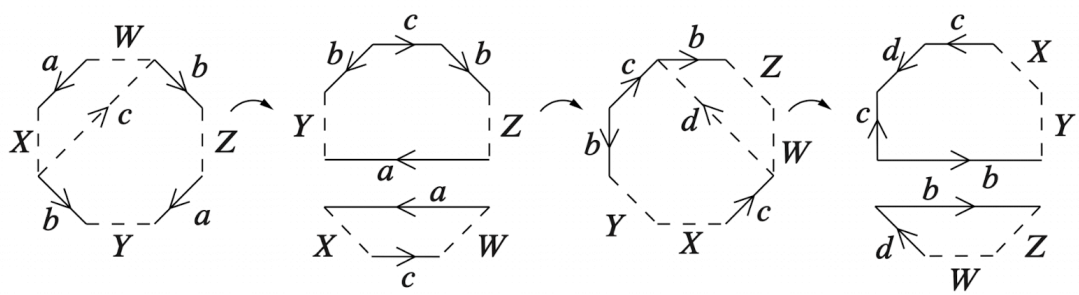

Fig. 6.22: Bringing intertwined complementary pairs together.

Conclusion

We proved that one presentation only contains blocks in the form $aba^{-1}b^{-1}$ or cc, and their identifiers do not repeat by Step2&3. Hence M is either $\mathbb{S}^2$, the connected sum of $\mathbb{T}^2$ or the connected sum of $\mathbb{P}^2$.

Part3: Conway's proof

We now introduce Conway's proof. Recall we proved rather easily that $K \cong \mathbb{P}^2 \# \mathbb{P}^2$ and $\mathbb{T}^2 \# \mathbb{P}^2 \cong \mathbb{P}^2 \# \mathbb{P}^2 \# \mathbb{P}^2$.
We call M an **ordinary surface**, if each component of M is a sphere with perforations (holes), along with attached tori and/or projective planes. Note that we can also attach a Klein bottle since $K \cong \mathbb{P}^2 \# \mathbb{P}^2$.
Define a **zip** to be a closed subset I of ∂M, homeomorphic to $[0, 1]$ or $\mathbb{S}^1$.
We only need to prove the following lemma:

Lemma: Suppose M is an ordinary surface (with boundary), and $I_1, I_2 \subset \partial M$ be disjoint zips. If $f : I_1 \to I_2$ is a homeomorphism, $M' = M/\sim$ is still an ordinary surface where $x \sim f(x)$.
Proof: We discuss the types of I_1, I_2:
Case1: If I_1, I_2 are entire boundary circles, homeomorphic to $\mathbb{S}^1$.
If they belong to different components of M, then glue them together is just a connected sum, so M' is still ordinary.
Otherwise, pasting I_1, I_2 adds either a handle($\#\mathbb{T}^2$) or a crosscap($\#K$) depending on whether f is orientation preserving or reversing. Hence M' is still ordinary.
Case2: If I_1, I_2 are arcs that partition the same boundary circle.
Then $I_1 \cup I_2 = C$ is homeomorphic to $\mathbb{S}^1$, and they only intersect at the two endpoints.
If f is orientation reversing, then M' is just M with the perforation removed.
Otherwise f joins I_1, I_2 with antipodal identification. This is just attaching a Möbius band to M, so $M' = M \# \mathbb{P}^2$ is ordinary.
Case3: If I_1, I_2 are arcs that leave gaps in their boundary circles.
First suppose I_1, I_2 are of different boundary circles C_1, C_2. By continuous deformation we may suppose $C_1 \backslash I_1$ and $C_2 \backslash I_2$ are microscopic. Then pasting I_1, I_2 together is just pasting C_1, C_2 together, and removing a tiny open disk formed by $C_1 \backslash I_1$ and $C_2 \backslash I_2$. By Case1 pasting C_1, C_2 makes an ordinary surface, and removing a perforation makes M' still an ordinary surface.
Likewise if I_1, I_2 are arcs of the same boundary circle, we can also suppose they take up almost all of the circle, then it is just pasting the circle as in Case2, and removing one or two open disks. Hence it also leaves M' as an ordinary surface.

Conclusion

We assume that any compact manifold M has a triangulation. Make each triangle into a sphere with a hole, where the edges are three arcs that partition the hole. Consider the ordinary surface M', consisting of these spheres, and for each edge in the triangulation, paste the spheres corresponding to the two faces containing the edge. By the lemma, we finally obtain an ordinary surface M'', which is exactly M. Hence if M is connected, it is $\mathbb{S}^2$, the connected sum of $\mathbb{T}^2$, or the connected sum of $\mathbb{P}^2$.